

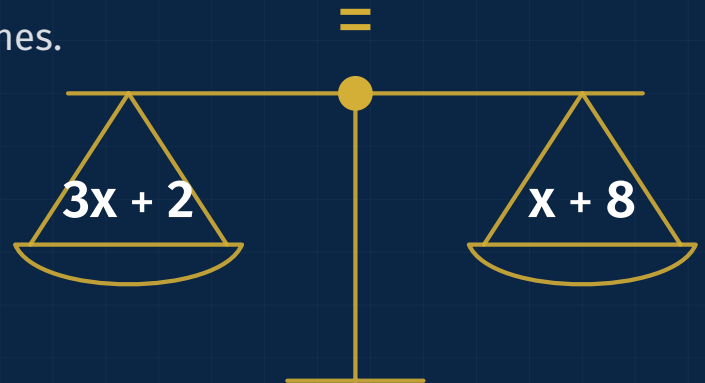


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CHAPTER 2 • ALGEBRA • TOPIC 2

Roots & Indices

Seven kinds of index question, one set of laws. Learn the laws, learn the power ladders, and every one is a few clean lines.



7

VARIANTS

60

PAST-PAPER PARTS

70

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Multiply & divide

V2 Brackets

V3 Zero & negative

V4 Fractional powers

V5 Flip → Root → Power

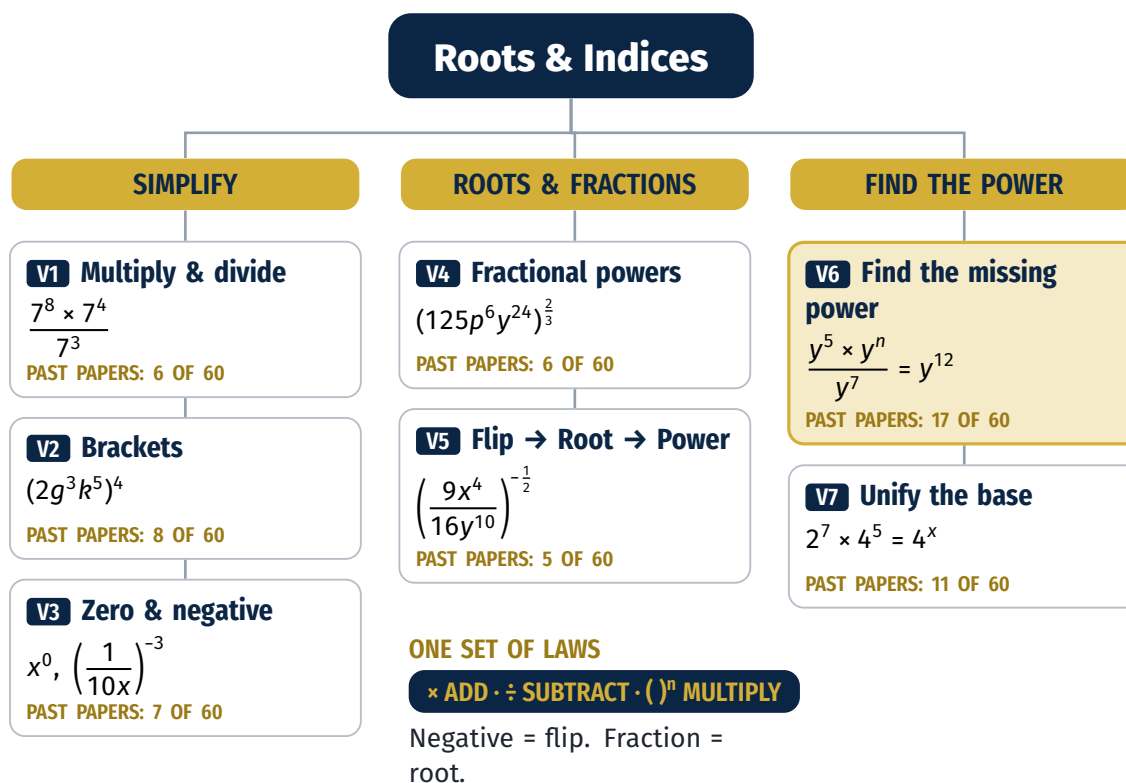
V6 Find the missing power

V7 Unify the base

TOPIC MAP · EVERY VARIANT ON ONE PAGE

Roots & Indices at a glance

 Your notes



★ How often does each variant come up?

33 past-paper questions (Jan 2020 → Nov 2025) hold 60 index parts; 70 practice parts — grouped by type.

	PAST PAPERS (OUT OF 60)	PRACTICE PARTS
V1 Multiply & divide	6	24
V2 Brackets	8	14
V3 Zero & negative	7	8
V4 Fractional powers	6	6
V5 Flip → Root → Power	5	5
V6 Find the missing power	17	7
V7 Unify the base	11	6

WHAT THIS TELLS YOU Almost every paper has an index question. “Find the value of n ” (V6 + V7) is 28 of the 60 parts — and the hardest marks.

BEFORE YOU START

The big idea

**A power is repeated multiplying.**

2^5 means five 2s multiplied. The 2 is the **base**, the 5 is the **index** (power).

Every law below is just **counting** how many of the base you have.

INDEX → 5
BASE → 2

$$2^5 = \underbrace{2 \times 2 \times 2 \times 2 \times 2}_{\text{FIVE 2s}} = 32$$

The four laws (Dr Eslam: $\times$ add $\cdot$ subtract $\cdot$ $()^n$ multiply)
Multiply → add the powers

$$a^3 \times a^2 = a^{3+2} = a^5$$

$(a \ a \ a)(a \ a)$ is five a 's.

Divide → subtract the powers

$$a^5 \div a^2 = a^{5-2} = a^3$$

Two a 's on top cancel with two below.

Bracket → multiply the powers

$$(a^3)^2 = a^{3 \times 2} = a^6$$

$a^3 \times a^3$: two lots of three.

Zero, negative, fraction

$$a^0 = 1, \quad a^{-n} = \frac{1}{a^n}, \quad a^{\frac{1}{n}} = \sqrt[n]{a}$$

Zero → 1. Negative → flip. Fraction → root.

Power ladders — learn them

BASE 2	2^1 2	2^2 4	2^3 8	2^4 16	2^5 32	2^6 64	2^7 128	2^8 256
BASE 3	3^1 3	3^2 9	3^3 27	3^4 81	3^5 243	3^6 729		
BASE 5	5^1 5	5^2 25	5^3 125	5^4 625				

CALCULATOR FIRST, THEN CHECK

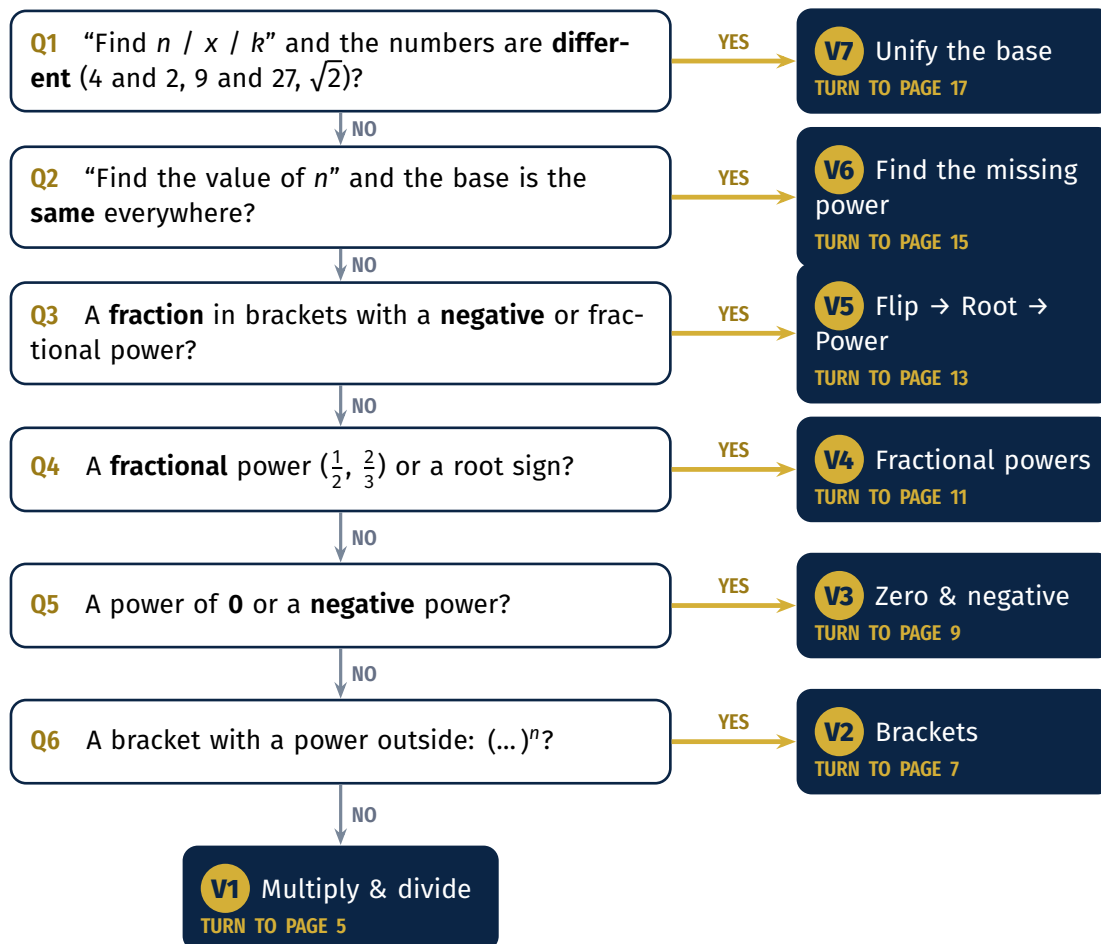
Test a law with numbers: $2^8 \div 2^3$ on the calculator is 32, and $2^5 = 32$ too. Use the $x^{\square}$ key and **SHIFT** $x^{\square}$ for roots. A fractional power: type $8 \ x^{\square} \ 2 \div 3 = 4$.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD



What is it?



How do I do it?



Worked example



Check yourself



When you see → you do



Watch out



Exam tip



Calculator check

V1

VARIANT 1

Multiply & Divide Powers

Same base: multiply → add the powers,
divide → subtract them.

SEEN IN PAST PAPERS

6 / 60

PRACTICE QUESTIONS: 24

 Your notes

Came up in: Jan 2020 P2H Q1(a)(b) · Jan 2022 P1H Q1(a)(b) · May 2023 P1HR Q6(a) · Nov 2023 P2H Q4(b)

? What is it?

In plain words

Powers with the **same base** are multiplied or divided. Count the letters: multiplying **adds** them, dividing **cancels** them.

Numbers in front are multiplied/divided as normal:
 $2a^3b \times 5a^2b^3 = 10a^5b^4$.

IT LOOKS LIKE

$$a^7 \times a^4$$

$$\frac{x^9}{x^2}$$

$$\frac{7^8 \times 7^4}{7^3}$$

$$a \ a \ a \times a \ a = a^5$$

$$\frac{a \ a \ a \ a \ a}{a \ a} = a^3$$

Multiply: 3 + 2 = 5 a's. Divide: 2 on top cancel with 2 below.

How do I do it?

1 Numbers first

Multiply or divide the numbers in front: $45 \div 5 = 9$.

2 $\times \rightarrow$ add

Same letter multiplied: **add** the powers.

3 $\div \rightarrow$ subtract

Same letter divided: **subtract** top power – bottom power.

4 Keep the base

The base never changes: $7^8 \times 7^4 = 7^{12}$, not 49^{12} .

$\times$ ADD $\div$ SUBTRACT Only for the same base.

V1 Multiply & Divide Powers

VARIANT 1 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO"Simplify $a^7 \times a^4$ "→ Add the powers: a^{11} "Write $\frac{7^8 \times 7^4}{7^3}$ as a single power of 7"

→ Keep base 7; add, then subtract

"Simplify $\frac{45e^6f^8}{5ef^2}$ "

→ Numbers divide; each letter subtracts

**WORKED EXAMPLE** **PAST PAPER**

January 2020 · Paper 2H · Q1(b) · 2 marks

Write $\frac{7^8 \times 7^4}{7^3}$ as a single power of 7.**STEP 2**

$$7^8 \times 7^4 = 7^{8+4} = 7^{12}$$

◀ multiply: add the powers

STEP 3

$$\frac{7^{12}}{7^3} = 7^{12-3}$$

◀ divide: subtract the powers

STEP 4

$$7^9$$

◀ keep the base 7

ANSWER 7^9

✓ calculator: both sides = 40 353 607

**WATCH OUT**

$$\times 7^8 \times 7^4 = 49^{12}$$

$$\checkmark 7^8 \times 7^4 = 7^{12}$$

You are counting 7s — the base stays 7.
Only the powers change.

**EXAM TIP**

Not sure? Test with small numbers on the calculator: $2^3 \times 2^2 = 32 = 2^5$.

**Check yourself**

calculator first · answers on page 19

1.1

Jan 2022 P1H

Simplify $a^7 \times a^4$

Answer:

1.2

May 2023 P1HR

Simplify $m^{10} \div m^3$

Answer:

1.3

Practice · Medium

Simplify $\frac{45e^6f^8}{5ef^2}$

Answer:

V2

VARIANT 2

Powers of Brackets

A power outside a bracket hits everything inside — the number too.

SEEN IN PAST PAPERS

8 / 60

PRACTICE QUESTIONS: 14



Your notes

Came up in: Jan 2021 P2HR Q4(a) • Jan 2022 P1H Q1(c) • Jun 2022 P1H Q9(c) • May 2023 P1HR Q6(c) • Nov 2023 P2H Q4(c) • May 2024 P1H Q10(c) • Jun 2025 P1H Q4(c) • Jun 2025 P2HR Q6(c)

? What is it?**In plain words**

$(x^3)^2$ means $x^3 \times x^3$: two lots of three, so the powers **multiply**: x^6 .

With more inside, the outside power goes to **every part** — the number is raised to the power, each letter's power is multiplied.

IT LOOKS LIKE

$$(8x^5y^3)^2$$

$$(2k^2m^4)^3$$

$$(5a^4c^2)^3$$

$$(2g^3k^5)^4$$

$$2^4 g^{3 \times 4} k^{5 \times 4} = 16g^{12}k^{20}$$

The power 4 goes to the 2, to g^3 and to k^5 .

How do I do it?**1 Every part**

The outside power goes to the number and to each letter.

2 The number

Raise it to the power: $2^4 = 16$, $5^3 = 125$ (calculator).

3 The letters

Multiply each power by the outside power: $(g^3)^4 = g^{12}$.

THE NUMBER GETS THE POWER TOO

Dr Eslam's rule for brackets.

V2 Powers of Brackets

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Simplify $(8x^5y^3)^2$ " → $8^2 = 64$; **powers** $\times 2$: $64x^{10}y^6$

"Simplify fully $(5a^4c^2)^3$ " → $5^3 = 125$; **powers** $\times 3$

$(m^4)^3$ or $(k^3)^2$ → **Multiply the powers**

**WORKED EXAMPLE** **PAST PAPER**

November 2023 · Paper 2H · Q4(c) · 2 marks

Simplify fully $(2g^3k^5)^4$

STEP 1 $2^4 \times (g^3)^4 \times (k^5)^4$ ◀ the 4 goes to every part

STEP 2 $2^4 = 16$ ◀ the number: $2 \times 2 \times 2 \times 2$

STEP 3 $g^{12} k^{20}$ ◀ letters: multiply the powers

ANSWER $16g^{12}k^{20}$ ✓ $g = k = 1$: both sides 16**WATCH OUT**

$$\times (2g^3k^5)^4 = 2g^{12}k^{20}$$

$$\checkmark (2g^3k^5)^4 = 16g^{12}k^{20}$$

The 2 is inside the bracket, so it is raised to the power 4 as well.

**EXAM TIP**

Work the number out on the calculator ($5^3 = 125$) — the marks are for the full answer, not $5^3a^{12}c^6$.

**Check yourself**

calculator first · answers on page 19

2.1

Jan 2022 P1H

Simplify $(8x^5y^3)^2$

Answer:

2.2

Jun 2022 P1H

Simplify fully $(2k^2m^4)^3$

Answer:

2.3

Practice · Hard

$(ar^b)^4 = 16r^{20}$, a, b
positive integers. Find
 a and b .

Answer:

V3

VARIANT 3

Zero & Negative Powers

Power 0 means 1. A negative power means flip it.

SEEN IN PAST PAPERS

7 / 60

PRACTICE QUESTIONS: 8



Your notes

Came up in: Jan 2021 P2HR Q4(b) · Jan 2022 P2HR Q6(a) · Jun 2022 P1H Q9(a) · Jun 2023 P2HR Q7(a) · Jun 2023 P2H Q15(b) · May 2024 P1H Q10(a) · Jun 2025 P2HR Q6(a)

? What is it?

In plain words

Go down the ladder: each step down **divides by the base**. So $2^0 = 1$ and $2^{-1} = \frac{1}{2}$, $2^{-2} = \frac{1}{4}$.

Zero power → 1 (anything except 0).

Negative power → flip: $a^{-n} = \frac{1}{a^n}$.

IT LOOKS LIKE

x^0 $8 \times (4t)^0$ 2^{-3} $\left(\frac{1}{10x}\right)^{-3}$

$$2^3 = 8$$

$$2^2 = 4$$

$$2^1 = 2$$

$$2^0 = 1$$

$$2^{-1} = \frac{1}{2}$$

$$2^{-2} = \frac{1}{4}$$

÷2
÷2
÷2
÷2
÷2

How do I do it?

1 Power 0

Whatever is under the 0 becomes 1: $(4t)^0 = 1$, so $8 \times (4t)^0 = 8$.

2 Negative → flip

Flip the base and make the power positive: $\left(\frac{1}{10x}\right)^{-3} = (10x)^3$.

3 Use the laws

Finish with V1/V2: $(10x)^3 = 1000x^3$.

NEGATIVE MEANS FLIP

Never “make it negative”: $2^{-3} = \frac{1}{8}$, not -8 .

V3 Zero & Negative Powers

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Write down the value of x^0 ” / 5^0 → **1**

“Simplify $8 \times (4t)^0$ ” → **Only the bracket is 1: $8 \times 1 = 8$**

“Express ... in the form px^n ” with a negative power → **Flip, then the bracket law**

**WORKED EXAMPLE****PAST PAPER**

June 2023 · Paper 2H · Q15(b) · 2 marks

Express $\left(\frac{1}{10x}\right)^{-3}$ in the form px^n where p and n are integers.

STEP 2

$$\left(\frac{10x}{1}\right)^3 = (10x)^3$$

◀ negative power: flip, power positive

STEP 3

$$10^3 x^3$$

◀ the 3 goes to every part

STEP 3

$$1000x^3$$

◀ $10^3 = 1000$

ANSWER $1000x^3$ ($p = 1000, n = 3$)

$$\checkmark x = 1: \left(\frac{1}{10}\right)^{-3} = 1000$$

**WATCH OUT**

$$\times 2^{-3} = -8$$

$$\checkmark 2^{-3} = \frac{1}{2^3} = \frac{1}{8}$$

A negative power never makes the answer negative — it makes it a fraction.

**EXAM TIP**

$5y^0 = 5$ (only y has the 0), but $(5y)^0 = 1$.
Look where the bracket is.

**Check yourself**

calculator first · answers on page 19

3.1

Jun 2022 P1H

Simplify $8 \times (4t)^0$

Answer:

3.2

Jan 2021 P2HR

Simplify
 $(21m^4n) \div (3n^{-5})$

Answer:

3.3

Practice · Medium

Work out the value of
 2^{-3}

Answer:

V4

VARIANT 4

Fractional Powers

A fraction as a power: the bottom is a root,
the top is a power.

SEEN IN PAST PAPERS

6 / 60

PRACTICE QUESTIONS: 6


 Your notes

Came up in: Jan 2021 P1HR Q12(a) • Jun 2023 P2HR Q11 • Jun 2023 P2H Q15(a) • Nov 2023 P2H Q12(b) • May 2024 P1HR Q17(a) • Nov 2025 P2H Q13(a)

? What is it?

In plain words

$a^{\frac{1}{2}} = \sqrt{a}$, $a^{\frac{1}{3}} = \sqrt[3]{a}$. In $a^{\frac{m}{n}}$ the
bottom n is the root and the
top m is the power.

Do the **root first** — the numbers
 stay small.

IT LOOKS LIKE

$$(16e^{10}f^6)^{\frac{1}{2}}$$

$$(32a^{15})^{\frac{3}{5}}$$

$$(125p^6y^{24})^{\frac{2}{3}}$$

$$8^{\frac{2}{3}}$$

TOP = POWER
 BOTTOM = ROOT

$$\sqrt[3]{8} = 2 \quad \text{THEN} \quad 2^2 = 4$$

ROOT FIRST

How do I do it?

1 Root (bottom)

Take the root of every part: number root; letter powers ÷ the bottom.

2 Power (top)

Raise every part to the top: number to the power; letter powers × the top.

3 Tidy

Write the number first, then the letters in order.

BOTTOM = ROOT · TOP = POWER

Letters: just multiply the power by the fraction.

V4 Fractional Powers

VARIANT 4 · SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

A power like $\frac{1}{2}$, $\frac{3}{5}$, $\frac{2}{3}$

→ Bottom = root, top = power

 $\left(\sqrt[4]{k^{12}}\right)^5 = k^n$ → A root is a fraction power: $k^{12 \times \frac{1}{4} \times 5}$

"Simplify fully"

→ Number worked out ($\sqrt[3]{125} = 5$), letters with one power each

WORKED EXAMPLE PAST PAPER

November 2023 · Paper 2H · Q12(b) · 2 marks

Simplify fully $(125p^6y^{24})^{\frac{2}{3}}$

STEP 1

$$\sqrt[3]{125} p^{6 \div 3} y^{24 \div 3} = 5p^2y^8$$

◀ root first (the bottom 3)

STEP 2

$$(5p^2y^8)^2$$

◀ now the power (the top 2)

STEP 3

$$25p^4y^{16}$$

◀ $5^2 = 25$, powers $\times 2$ ANSWER $25p^4y^{16}$ ✓ $p = y = 1$: $125^{2/3} = 25$ 

WATCH OUT

$$\times 125^{\frac{2}{3}} = 125 \times \frac{2}{3}$$

$$\checkmark 125^{\frac{2}{3}} = (\sqrt[3]{125})^2 = 25$$

A fraction in the power is a root and a power — never multiply the number by it.



EXAM TIP

For the letters it is quicker: multiply the power by the fraction. $y^{24 \times \frac{2}{3}} = y^{16}$.



Check yourself

calculator first · answers on page 19

4.1

Jan 2021 P1HR

Simplify $(16e^{10}f^6)^{\frac{1}{2}}$

Answer:

4.2

Jun 2023 P2H

Simplify fully $(32a^{15})^{\frac{3}{5}}$

Answer:

4.3

May 2024 P1HR

 $\left(\sqrt[4]{k^{12}}\right)^5 = k^n$. Find n .

Answer:

V5

VARIANT 5

Flip → Root → Power

A fraction in brackets with a negative or fractional power: three moves, in order.

SEEN IN PAST PAPERS

5 / 60

PRACTICE QUESTIONS: 5



Your notes

Came up in: Jan 2020 P1H Q17(b) · Jan 2021 P2H Q11 · Jan 2021 P2HR Q19(b) · May 2021 P2H Q14 · Jun 2025 P2H Q15

? What is it?

IN PLAIN WORDS

A whole fraction is in a bracket, and the power is **negative** (and maybe a fraction). Do it in **three moves**: **flip** the fraction (the power becomes positive), take the **root** (bottom of the power), then the **power** (top).

IT LOOKS LIKE

$$\left(\frac{27a^{12}}{t^{15}}\right)^{-\frac{2}{3}} \quad \left(\frac{9x^4}{16y^{10}}\right)^{-\frac{1}{2}} \quad \left(\frac{2a^3}{10a^7c^2}\right)^{-3}$$

1 FLIP**2 ROOT****3 POWER**

$$\left(\frac{16y^{10}}{9x^4}\right)^{\frac{1}{2}}$$

$$\frac{4y^5}{3x^2}$$

$$\left(\frac{4y^5}{3x^2}\right)^1$$

$$\left(\frac{9x^4}{16y^{10}}\right)^{-\frac{1}{2}}$$

Dr Eslam's three moves. Each move is one line of working — and one mark.

How do I do it?

1 Flip

Turn the fraction upside down; the power loses its minus sign.

2 Root

The bottom of the power: root every number, divide every letter power.

3 Power

The top of the power: raise every part (if it is 1, nothing to do).

4 Simplify inside

Tip: if the bracket has the same letter top and bottom, simplify it **first**.

FLIP → ROOT → POWER Always in this order.

V5 Flip → Root → Power

VARIANT 5 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

“Simplify fully $\left(\frac{\dots}{\dots}\right)^{-\frac{2}{3}}$ ”

→ **Flip → Root → Power**

Same letter top and bottom: $\frac{2a^3}{10a^7c^2}$

→ **Simplify inside first:** $\frac{1}{5a^4c^2}$

“ $n^{-\frac{4}{5}} = \left(\frac{1}{2}\right)^4$, find n ”

→ **Undo: power $-\frac{5}{4}$ on both sides****WORKED EXAMPLE** **PAST PAPER**

May 2021 · Paper 2H · Q14 · 3 marks

Simplify fully $\left(\frac{9x^4}{16y^{10}}\right)^{-\frac{1}{2}}$

STEP 1

$$\left(\frac{16y^{10}}{9x^4}\right)^{\frac{1}{2}}$$

◀ flip: the power is now $+\frac{1}{2}$ **STEP 2**

$$\frac{\sqrt{16}y^{10 \div 2}}{\sqrt{9}x^{4 \div 2}}$$

◀ root: the bottom 2 is a square root

STEP 3

$$\frac{4y^5}{3x^2}$$

◀ power: the top is 1 — done

ANSWER $\frac{4y^5}{3x^2}$

$$\checkmark x = y = 1: \left(\frac{9}{16}\right)^{-\frac{1}{2}} = \frac{4}{3}$$

**WATCH OUT**

$$\times \left(\frac{9}{16}\right)^{-\frac{1}{2}} = -\frac{3}{4}$$

$$\checkmark \left(\frac{9}{16}\right)^{-\frac{1}{2}} = \frac{4}{3}$$

The minus in the power means flip — the answer is not negative.

**EXAM TIP**

Write three lines: flipped, rooted, powered. Each one is a method mark.

**Check yourself**

calculator first · answers on page 19

5.1

Jan 2020 P1H

Simplify fully
 $\left(\frac{27a^{12}}{t^{15}}\right)^{-\frac{2}{3}}$

Answer:

5.2

Jun 2025 P2H

Simplify fully
 $\left(\frac{2a^3}{10a^7c^2}\right)^{-3}$

Answer:

5.3

Jan 2021 P2H

Simplify fully
 $\left(\frac{9t^4w^9}{18t^6w^{10}}\right)^{-2}$

Answer:

V6**VARIANT 6** MOST COMMON IN THE EXAM

Find the Missing Power

Same base on both sides: make each side one power, then the powers are equal.

SEEN IN PAST PAPERS

17 / 60

PRACTICE QUESTIONS: 7



Your notes

Came up in: Jan 2022 P2HR Q6(b)(c) • Jun 2022 P1H Q9(b) • Jun 2023 P2HR Q7(b) • Jun 2024 P2HR Q7 • Jun 2025 P1H Q4(a)(b) • Jun 2025 P2HR Q6(b) • May 2021 P1H Q11(b) • May 2023 P1HR Q6(b) • May 2024 P1H Q10(b) • Nov 2021 P2H Q1(a)(b)(c) • Nov 2023 P2H Q4(d) • Nov 2024 P2H Q14 • Nov 2025 P1H Q8

? What is it?

In plain words

An equation where the unknown is **in a power** and every term has the **same base**.

Use the laws to make each side **one power** of that base. If $y^{\square} = y^{12}$, the boxes must be equal — that is a normal equation.

IT LOOKS LIKE

$$\frac{y^5 \times y^n}{y^7} = y^{12}$$

$$7^8 \times 7^2 \div 7^p = 7^6$$

$$\frac{y^5 \times y^n}{y^7} = y^{12}$$



ONE POWER EACH SIDE

$$y^{5+n-7} = y^{12}$$



SAME BASE → SAME POWER

$$n - 2 = 12$$

How do I do it?

1 One power each side

Use $\times$ add, $\div$ subtract, $()^n$ multiply until each side is base $^{\square}$.

2 Equate the powers

Same base, so the powers are equal: write that equation.

3 Solve

Solve the small linear equation for n .

4 Check

Put n back: same power on both sides?

SAME BASE → SAME POWER

The most common index question on the paper.

V6 Find the Missing Power

VARIANT 6 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Given that ... = y^n , find the value of n ” → **Simplify to one power, equate**

“Write down the value of m ”,
 $3^4 \times 3^5 = 3^m$ → **1 mark: just add, $m = 9$**

n appears twice: $3^{2n+3} \div 3^4 = 3^3 \times 3^{1-2n}$ → **Equate the powers, solve the linear equation**

WORKED EXAMPLE PAST PAPER

November 2023 · Paper 2H · Q4(d) · 2 marks

Given that $\frac{y^5 \times y^n}{y^7} = y^{12}$, work out the value of n .

- | | | |
|---------------|----------------------|---|
| STEP 1 | $y^{5+n-7} = y^{12}$ | ◀ one power on the left: add, then subtract |
| STEP 1 | $y^{n-2} = y^{12}$ | ◀ simplify the power |
| STEP 2 | $n - 2 = 12$ | ◀ same base $\Rightarrow$ same power |
| STEP 3 | $n = 14$ | ◀ solve |

ANSWER $n = 14$ $\checkmark 5 + 14 - 7 = 12$ **WATCH OUT**

✗ $y^5 \times y^n = y^{5n}$

✓ $y^5 \times y^n = y^{5+n}$

Multiplying powers of the same base **adds** the indices — even when one is a letter.

EXAM TIP

17 of the 60 past-paper index parts are this variant. Write the line “ $n - 2 = 12$ ”: it shows the examiner your method.

Check yourself

calculator first · answers on page 19

6.1 Nov 2021 P2H

$$\frac{7^8 \times 7^2}{7^p} = 7^6. \text{ Find } p.$$

Answer:

6.2 Jun 2024 P2HR

$$\frac{3^{-2} \times 3^5}{3^{10}} = 3^n. \text{ Find } n.$$

Answer:

6.3 Nov 2024 P2H

$$\frac{3^{2n+3}}{3^4} = 3^3 \times 3^{1-2n}. \text{ Find } n.$$

Answer:

V7

VARIANT 7

Unify the Base

Different numbers (4, 8, 16, $\sqrt{2}$): rewrite them all as powers of one prime.

SEEN IN PAST PAPERS

11 / 60

PRACTICE QUESTIONS: 6

 Your notes

Came up in: Jan 2020 P2HR Q21(b) • Oct/Nov 2020 P1HR Q21 • Jan 2021 P1HR Q12(c) • Jan 2021 P2HR Q19(a) • May 2021 P1H Q11(a) • Jun 2022 P2HR Q7 • Jun 2022 P2H Q24 • May 2023 P1HR Q24 • Nov 2023 P2H Q12(a) • Jun 2024 P2H Q23 • Nov 2025 P2H Q13(b)

? What is it?**In plain words**

The bases are **different** numbers — but they are powers of the same prime: $4 = 2^2$, $8 = 2^3$, $16 = 2^4$, $\sqrt{2} = 2^{\frac{1}{2}}$.

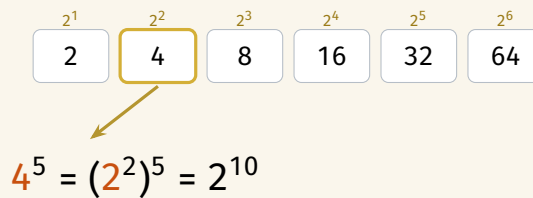
Unify the base: rewrite everything with that prime. Then it is V6.

IT LOOKS LIKE

$$2^7 \times 4^5 = 4^x$$

$$\sqrt{2} \times 16 = 2^x$$

$$4^{k+3} = 16 \times 2^k$$

2

SAME BASE — NOW USE THE LAWS

How do I do it?**1 Unify the base**

Write every number as a power of one prime (use the ladders on page 3).

2 Bracket law

$(2^2)^5 = 2^{10}$, $(2^2)^x = 2^{2x}$ — multiply the powers.

3 One power each side

$\times$ add, $\div$ subtract, as in V6.

4 Equate + solve

Same base $\Rightarrow$ same power. Solve for the letter.

UNIFY THE BASE Dr Eslam's rule: one prime, then compare powers.

V7 Unify the Base

VARIANT 7 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO
 $4, 8, 16, 32, 64$ or $\sqrt{2}$ with a power of 2 → **Base 2:** $4 = 2^2, 8 = 2^3, \sqrt{2} = 2^{\frac{1}{2}}$
 $9, 27, 81, \sqrt{27}$ with powers of 3 → **Base 3:** $9 = 3^2, 27 = 3^3$

 “Express x in terms of n ” → **Same method — the answer has n in it**
WORKED EXAMPLE **PAST PAPER**

November 2023 · Paper 2H · Q12(a) · 2 marks

 $2^7 \times 4^5 = 4^x$. Calculate the value of x .

STEP 1 $2^7 \times (2^2)^5 = (2^2)^x$

◀ unify: every 4 is 2^2

STEP 2 $2^7 \times 2^{10} = 2^{2x}$

◀ bracket law: multiply the powers

STEP 3 $2^{17} = 2^{2x}$

◀ × add: $7 + 10$

STEP 4 $17 = 2x \Rightarrow x = 8.5$

◀ same base → same power, solve

ANSWER $x = 8.5$ ✓ calculator: $2^7 \times 4^5 = 4^{8.5} = 131\,072$ **WATCH OUT**

$\times 4^5 = 2^5$

$\checkmark 4^5 = (2^2)^5 = 2^{10}$

 When you swap 4 for 2^2 , the power gets multiplied by 2.
EXAM TIP
 11 of the 60 parts — often the last index question on Paper 2. Learn the ladders: 2, 4, 8, 16, 32, $64 \cdot 3$, 9, 27, $81 \cdot 5$, 25, 125.
Check yourself

calculator first · answers on page 19

7.1

May 2021 P1H

$\sqrt{2} \times 16 = 2^x$. Find x .

Answer:

7.2

Jan 2021 P2HR

 Simplify $8^2 \times \sqrt[3]{4^6}$. Give the answer as 2^a .

Answer:

7.3

Jan 2021 P1HR

$4^{k+3} = 16 \times 2^k$. Find k .

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py). Got one wrong? Go back to that variant's Watch-out box first.



Your notes

1.1 a^{11}

1.2 m^7

1.3 $9e^5f^6$

2.1 $64x^{10}y^6$

2.2 $8k^6m^{12}$

2.3 $a = 2, b = 5$

3.1 8

3.2 $7m^4n^6$

3.3 $\frac{1}{8}$

4.1 $4e^5f^3$

4.2 $8a^9$

4.3 $n = 15$

5.1 $\frac{t^{10}}{9a^8}$

5.2 $125a^{12}c^6$

5.3 $4t^4w^2$

6.1 $p = 4$

6.2 $n = -7$

6.3 $n = \frac{5}{4}$

7.1 $x = 4.5$

7.2 2^{10}

7.3 $k = -2$

THE WHOLE TOPIC ON ONE PAGE

Roots & Indices cheat sheet



Your notes
V1 Multiply & divide 6/60

$$a^7 \times a^4 = a^{11}, \quad a^9 \div a^2 = a^7$$

× add, ÷ subtract. The base never changes.

V2 Brackets 8/60

$$(2g^3k^5)^4 = 16g^{12}k^{20}$$

The number gets the power too; letter powers multiply.

V3 Zero & negative 7/60

$$a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$$

Zero → 1. Negative → flip (never a negative answer).

V4 Fractional 6/60

$$(125p^6y^{24})^{\frac{2}{3}} = 25p^4y^{16}$$

Bottom = root, top = power. Root first.

V5 Flip → Root → Power 5/60

$$\left(\frac{9x^4}{16y^{10}}\right)^{-\frac{1}{2}} = \frac{4y^5}{3x^2}$$

Three moves, three lines.

V6 Find the power 17/60

$$y^{n-2} = y^{12} \Rightarrow n = 14$$

One power each side, then equate the powers.

V7 Unify the base 11/60

$$2^7 \times (2^2)^5 = 2^{27}$$

4, 8, 16, $\sqrt{2} \rightarrow$ base 2. Then V6.

**TOP 3 LOST MARKS**

1. $(2g^3)^4 = 2g^{12} \times$
2. $2^{-3} = -8 \times$
3. $4^5 = 2^5 \times$

BOTTOM LINE × add · ÷ subtract · ()ⁿ multiply · negative = flip · fraction = root · to solve: unify the base, equate the powers.